

Adaptive Risk-Aware Autonomy for Robotic Exploration of Uncertain Extraterrestrial Terrain

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Abstract

A planetary rover cannot be driven in real time from Earth, so it must decide for itself which ground is safe enough to cross. Those decisions rest on a terrain model built before launch, from whatever experience was available at the time. This work asks what happens when that model is wrong — when the robot arrives somewhere whose ground does not behave the way its model says — and whether a robot that revises its model from its own driving survives better than one that does not.

We build a controlled 2.5-D autonomy testbed in which a single solar-powered rover must visit science targets and return to its lander before its energy runs out, on terrain where wheel slip is stochastic, sensing is noisy and range-limited, and sustained slip ends the mission by embedding the vehicle. Three planners are compared on identical terrain: distance-only A*, a fixed risk-aware planner, and an otherwise identical planner whose mobility model is updated online from measured slip. The three differ only in their objective and in whether that objective’s inputs are revised, so a measured difference is attributable to the decision rule rather than to a different search implementation.

The central experiment is a domain shift. Every planner carries a mobility prior calibrated on lunar terrain and is then evaluated on Martian terrain, where the same terrain-class vocabulary carries substantially different slip statistics. Across 750 missions on frozen, held-out seeds, we report the effect of online adaptation on mission success and science return, and the generalization gap between in-distribution and out-of-distribution performance.

This is a simulation study of decision-making, not a validated model of any flight vehicle or landing site. Section 8 states plainly what the model does and does not represent.

1 Introduction

Round-trip light time to Mars runs from roughly eight to forty minutes, and contact windows are intermittent. A surface robot therefore cannot be joysticked around a hazard; by the time an operator sees a problem, the problem has already happened. Autonomy on planetary rovers exists for this reason, and the quality of that autonomy is measured by what the robot does when nobody can help it.

The failure mode that actually strands planetary rovers is not driving off a cliff. It is sinking. A rover crossing ground that looks benign can lose traction in loose material, dig itself in, and become permanently immobile. NASA’s *Spirit* ended its driving mission this way in 2009, embedded in soft soil. What makes this failure mode interesting for autonomy research is that it is invisible to geometry: the ground that traps a rover is often flat. Detecting it requires a model of how terrain

behaves, not how it *looks*, and the only direct measurement of behaviour is slip observed while driving on it.

That creates an asymmetry this paper is built around. Geometry — slope, roughness, obstacles — can be sensed at a distance from imagery. Mobility can only be measured by committing the vehicle to the ground. A robot therefore plans long routes using a terrain model it cannot verify in advance, and the model it starts with was calibrated somewhere else.

Research question. Does a planetary robot that continuously updates its terrain-mobility model from its own driving make safer and more productive decisions than one using a fixed model, when the terrain it encounters does not match the model it was given?

Hypotheses. Stated before the confirmatory data were generated (see `docs/PREREGISTRATION.md`):

- H1** On in-distribution terrain, where the prior is already correct, adaptation neither helps nor harms: the adaptive and fixed planners perform equivalently.
- H2** On out-of-distribution terrain, the adaptive planner returns a higher paired science fraction than the fixed planner.
- H3** Under increased slip dispersion and injected faults, the adaptive planner achieves a higher mission-success rate and fewer severe-slip events.
- H4** Intervention delay reduces productivity for all planners, and most for those that request more interventions.

What is and is not claimed as novel. Risk-aware planetary path planning, learned traversability, uncertainty-aware terrain models and energy-constrained planning all exist in the literature and are cited in Section 2. This work does not claim to invent any of them. What it contributes is a controlled measurement: holding the search algorithm, the cost function, the mission logic and the terrain fixed, and changing *only* whether the mobility model is revised online, in order to isolate what that single mechanism is worth under a realistic domain shift.

2 Related work

Slip has long been treated as the operative traversability signal for Mars rovers rather than slope alone; Angelova et al. learned to predict slip from terrain appearance, and Helmick et al. built terrain-adaptive navigation around slip-aware models. More recent work makes the uncertainty explicit rather than producing a single traversability score: Endo et al. fuse probabilistic traversability predictions for risk-aware planning on heterogeneous terrain, and extend this with test-time adaptation so that a traversability model can be corrected during deployment. Sanchez-Ibanez et al. address a complementary risk, planning under localization uncertainty for space exploration rovers.

The distinction this paper draws — between uncertainty that more driving can reduce and variability that it cannot — follows the same motivation as that line of work. Our contribution is not a better traversability model but a controlled comparison of what online correction of such a model is worth when the deployment environment differs from the calibration environment.

3 Simulation model

3.1 Scope

The simulator is a controlled testbed for comparing decision rules. It is not a validated dynamics model of a specific rover, and its absolute outputs are properties of the model rather than predictions of on-surface performance. Its purpose is to expose different planners to *identical* stochastic worlds so that differences between them are attributable to their decisions.

3.2 Terrain

Terrain is a $N \times N$ grid with an elevation field $z(x)$, from which slope $\theta(x)$ is the gradient magnitude in degrees. Each cell carries a terrain class $k(x)$ from a five-member vocabulary shared by both bodies: smooth regolith, rocky, loose fines, bedrock, and rim talus. Each cell also carries a roughness value, an illumination fraction $I(x) \in [0, 1]$, and a hazard flag.

Crucially, the class *vocabulary* is shared across bodies while the class *parameters* are not. Each body assigns every class a true mean slip μ_k , dispersion σ_k , and locomotion multiplier m_k . This is the mechanism of the domain shift: a robot arriving at a new body recognizes the same terrain categories but the statistics behind them have changed.

3.3 Mobility and embedding

When the robot commands a move into cell x , the realized slip fraction is

$$s \sim \mathcal{N}\left(\mu_{k(x)} + \beta \theta(x) + b, \sigma_{k(x)}^2\right), \quad s \in [0, 0.995], \quad (1)$$

with $\beta = 0.01$ per degree, representing traction lost on grades, and b a bias introduced by wheel-damage faults. A step is *severe* when $s \geq \tau$ with $\tau = 0.80$: the wheels turn but the robot does not advance. After $L = 3$ consecutive severe steps the robot is permanently embedded and the mission ends. This is the model’s representation of the Spirit-style loss mode, and it is the only failure with no recovery.

3.4 Energy

Energy to traverse a cell is

$$E(x) = \frac{E_0 d m_{k(x)} (1 + \gamma \theta(x))}{\max(1 - s, \epsilon)} \cdot \frac{g}{g_{\text{ref}}} \cdot \frac{1}{\eta_m}, \quad (2)$$

with d the distance in cells, $\gamma = 0.035$ per degree, $\epsilon = 0.08$, g the body’s gravity against a lunar reference $g_{\text{ref}} = 1.62 \text{ m/s}^2$, and η_m a motor-efficiency factor degraded by faults. The slip term sits in the denominator because slipping wheels still consume power while delivering less ground: at slip s , covering the same distance costs roughly $1/(1 - s)$ times as much.

Equation 2 is the reason a mobility error is also an energy error, and it is the mechanism behind the results in Section 6. A robot that underestimates slip underestimates the cost of every route it considers.

Solar income per timestep is $\rho I(x) \eta_s$, with ρ the nominal harvest rate and η_s an efficiency degraded by dust faults. Battery capacity is scaled by g/g_{ref} , representing a vehicle whose power system is sized for the body it was built for; what transfers between bodies in this study is the autonomy, not the hardware.

3.5 Sensing

The robot never reads ground truth. Within a sensing radius it receives noisy estimates of slope, roughness, illumination, terrain class and hazard status, with noise growing linearly in range. Terrain class is misclassified with a range-dependent probability. Slip, by contrast, is measured directly and without class ambiguity — but only for the single cell just driven.

4 Autonomy stack

4.1 World model

The robot maintains a belief, not the terrain. Per cell it stores the most recent close-range geometric estimate. Per terrain class it stores a posterior over mean slip, updated from driven experience by a Normal–Normal conjugate model: with prior mean m_0 , prior variance t_0^2 , observation variance σ_o^2 and n measurements of sample mean $\bar{s}$,

$$\lambda = \frac{1}{t_0^2} + \frac{n}{\sigma_o^2}, \quad m_n = \frac{1}{\lambda} \left(\frac{m_0}{t_0^2} + \frac{n\bar{s}}{\sigma_o^2} \right), \quad t_n^2 = \frac{1}{\lambda}. \quad (3)$$

The posterior is recomputed from the prior and the full sample each time, so it does not depend on update order.

Generalizing to unseen ground. Routes are composed mostly of cells the robot has never observed, so what it believes about *unknown* terrain determines almost every plan. For an unobserved cell the class is unknown, and the expected slip is the current class beliefs marginalized over the class mix π_k actually encountered so far:

$$\hat{s}_{\text{unk}} = \sum_k \pi_k m_n^{(k)}. \quad (4)$$

This is what allows learning to transfer: measuring that the local drift sand is treacherous raises the robot’s estimate for ground it has not yet reached. A fixed constant in place of Equation 4 would confine every lesson to cells already visited and, because routes are mostly unvisited cells, would prevent adaptation from affecting any planning decision at all.

Two kinds of uncertainty. For unobserved cells the epistemic term combines the spread *between* class means with the uncertainty *within* each:

$$\sigma_{\text{epi}}^2 = \underbrace{\sum_k \pi_k (m_n^{(k)} - \hat{s}_{\text{unk}})^2}_{\text{between classes}} + \underbrace{\sum_k \pi_k t_n^2(k)}_{\text{within class}}, \quad \sigma_{\text{tot}}^2 = \sigma_{\text{epi}}^2 + \sigma_{\text{ale}}^2. \quad (5)$$

The separation matters: σ_{epi} is reducible by driving, whereas σ_{ale} is the terrain’s own variability and is not. A planner that cannot tell them apart cannot distinguish ground that is dangerous from ground that is merely unvisited.

4.2 Risk

Under the belief that slip at x is $\mathcal{N}(\hat{s}, \sigma_{\text{tot}}^2)$, the probability of a severe step and of embedding are

$$p_{\text{sev}}(x) = 1 - \Phi\left(\frac{\tau - \hat{s}(x)}{\sigma_{\text{tot}}(x)}\right), \quad p_{\text{emb}}(x) = p_{\text{sev}}(x)^L. \quad (6)$$

Treating consecutive steps as independent *understates* risk in homogeneous terrain, where severe slip is spatially correlated and runs are more likely than independence implies. The approximation is applied identically to every planner and so cannot favour one.

A believed hazard is not by itself mission-ending: the robot’s onboard geometric check refuses such a command and replans. It enters the risk only through the residual probability $\kappa = 0.005$ that the check misses it:

$$R(x) = 1 - (1 - p_{\text{emb}}(x))(1 - \kappa p_{\text{haz}}(x)), \quad R(\pi) = 1 - \prod_{x \in \pi} (1 - R(x)). \quad (7)$$

Energy risk is evaluated against the usable budget, treating required energy as normal with variance propagated from slip uncertainty by linearizing Equation 2 about $\hat{s}$:

$$\sigma_E^2 = \sum_{x \in \pi} \left(\frac{E(x)}{1 - \hat{s}(x)} \right)^2 \sigma_{\text{tot}}^2(x), \quad P_{\text{energy}} = \Pr[E_{\text{req}} \geq B + S - R_{\text{res}}], \quad (8)$$

with B the charge in hand, S the solar income expected over the route, and R_{res} a reserve held back as a fraction of capacity. Total mission failure probability composes the two causes.

4.3 Mission logic

A mission is: start at the lander, visit science targets, return before the energy runs out. The manager pursues the reachable target with the best value-to-cost ratio whose round trip satisfies the risk budget $P_{\text{fail}} \leq \varepsilon$, and otherwise returns home. Missions are flown as sorties — a target that does not fit on a low battery is reconsidered after recharging, and written off only when the robot is home, recharged, and still cannot reach it.

When no route is believed to exist, the robot must ask mission control. It idles for the communication delay before receiving a relaxed instruction, and each such episode is counted as a human intervention. An autonomy system that succeeds only by asking for help frequently has not really succeeded, so intervention count is reported alongside the outcome measures.

4.4 The three planners

All three run the same A* implementation over the same believed world and differ only in the cost assigned to entering a cell.

Distance-only A*

$J = d$. The control with no terrain model at all.

Fixed risk-aware A*

$J = w_d d + w_e E(x) + w_r R(x) + w_u \sigma_{\text{epi}}(x)$, consulting the world model but never revising it.

Adaptive risk-aware A*

Identical objective and identical weights; the single difference is that measured slip is folded into the class belief via Equation 3, so future plans use corrected estimates.

The adaptive planner is implemented as a subclass of the fixed planner that overrides one method. The experimental treatment is therefore a single switch, which is what makes the contrast interpretable.

5 Experimental design

5.1 Seed protocol

Terrain seeds are partitioned into disjoint integer ranges — train, validation, test and out-of-distribution — frozen to a checksummed file before any result was generated. Priors are calibrated on train seeds only. All tuning was performed on validation seeds. Test and OOD seeds informed no design decision.

Eight seeds consumed by an earlier engineering pilot are permanently quarantined and excluded from every confirmatory result; a regression test asserts they cannot reappear. The confirmatory run used 50 seeds per condition from split checksum `c9346c400ce5` at commit `a5e5081728c8`.

5.2 Design

3 planners $\times$ 5 conditions $\times$ 50 seeds = 750 missions. Every planner is evaluated on identical terrain within a condition, making this a randomized block design in which terrain seed is the block.

Conditions: Moon in-distribution; Mars out-of-distribution; Mars with $1.5\times$ slip dispersion; Mars with injected faults; and Mars with a 20-step intervention delay. Every condition uses a lunar-calibrated prior, so the Martian conditions are genuine domain shifts.

5.3 Analysis

Because planners share terrain within a condition, all contrasts are paired. Science fraction uses a paired t -test with a Wilcoxon signed-rank companion, Cohen’s d_z , and a 95% confidence interval. Mission success is binary and paired, so it uses McNemar’s exact test, which conditions on the seeds where the two planners actually disagreed. Holm–Bonferroni correction is applied across the prespecified primary family; secondary contrasts against the distance-only control are corrected separately and never pooled with it.

We also report the generalization gap $G = M_{\text{in}} - M_{\text{out}}$ per planner, a between-condition descriptive quantity reported without a paired test because the conditions draw from different seed splits.

6 Results

All 750 confirmatory missions completed without software error and none were excluded. Results are reported for every outcome regardless of significance, as the analysis plan requires.

6.1 In-distribution: adaptation is neutral (H1 supported)

On lunar terrain, where the prior is already correct, the adaptive and fixed risk-aware planners are statistically indistinguishable: success 0.92 versus 0.90 ($\Delta = +0.020$, $p_{\text{Holm}} = 1.000$), differing on only five of fifty seeds. This is what H1 predicted: when there is nothing to learn, learning costs nothing.

Both risk-aware planners clearly beat the distance-only control here (0.66 success, 0.659 science fraction), confirming that modelling terrain risk is worth doing when the model is right.

6.2 Out-of-distribution: H2 is not supported

H2 predicted that adaptation would raise science fraction on Martian terrain. It did not. In the baseline Martian condition the paired difference in science fraction was -0.015 ($p_{\text{Holm}} = 1.000$) — a point estimate in the *opposite* direction to the hypothesis — and mission success was exactly tied, with each planner winning five of the ten seeds on which they differed.

We report this as a failure of the stated hypothesis rather than reframing it. The validation-set estimate that motivated H2 (a success advantage of $+0.125$ to $+0.167$) did not replicate on held-out seeds. That divergence is itself the argument for holding seeds back: a tuning-set effect of that size vanished entirely under confirmatory evaluation.

6.3 Degraded conditions: H3 partially supported

The one contrast surviving Holm correction across the ten-member primary family is mission success under injected hardware faults, where adaptation raised success from 0.26 to 0.46 ($\Delta = +0.200$, $p_{\text{Holm}} = 0.020$). The discordant pairs are one-sided: of the ten seeds where the two planners disagreed, the adaptive planner succeeded on all ten and the fixed planner on none.

The other two degraded conditions move the same way without reaching corrected significance: $+0.140$ under elevated slip dispersion (12 of 13 discordant seeds favouring adaptation, $p_{\text{Holm}} = 0.461$) and $+0.160$ under intervention delay ($p_{\text{Holm}} = 0.461$). The direction is consistent across all three degraded conditions, which is suggestive, but only one is individually defensible after correction and we do not claim more.

6.4 Adaptation buys survival, not science

Across every Martian condition the sign pattern is the same: adaptation improves mission success while slightly *reducing* science fraction (for example -0.044 under intervention delay). The mechanism is visible in the termination counts. Under faults, energy exhaustion falls from 24 missions to 17, and ground interventions fall from 95.6 to 66.8 per mission.

This follows directly from Equation 2. A robot that has measured that the local ground is slipperier than its prior assumed raises its estimate of what every remaining route will cost, and therefore turns back sooner. Turning back sooner is exactly what converts a stranded rover into a surviving one — and exactly what costs it the last science target. The adaptive planner is not better at everything; it is more accurate about its own energy budget, and accuracy here purchases caution.

6.5 An unanticipated result: the uninformed planner survives

None of the hypotheses anticipated the clearest pattern in the Martian data. Distance-only A*, which models no terrain risk whatsoever, achieved the *highest* mission-success rate of the three planners in three of the four Martian conditions — 0.48 against 0.34 and 0.34 in the baseline OOD condition.

It also returned the least science (0.101 against 0.173 and 0.158). The interpretation we favour, which the intervention counts support, is that risk-averse routing on hostile terrain is expensive: the risk-aware planners detour around suspect ground, and on Mars almost all ground is suspect, so they travel further, request far more ground interventions (102.8 per mission against 57.5 for distance-only), and more often run the battery down before getting home. The short, blunt route is more dangerous per metre but involves fewer metres.

This is a genuine limitation of the risk formulation rather than a curiosity. A planner that treats caution as free will over-pay for it when the whole map is dangerous.

6.6 Generalization gap

The generalization gap G is largest for the adaptive planner (0.670) and smallest for distance-only A* (0.558). This ranking should not be read as adaptation generalizing worst. G is a difference, and the adaptive planner starts from the highest in-distribution score, so the metric mechanically rewards planners that are poor at home. Reported for completeness, it is uninformative for ranking here.

Table 1: Outcomes by condition and planner. Immob. and Energy out are counts of missions ending in embedding and in energy exhaustion respectively.

Condition	Planner	n	Success	Science frac.	Immob.	Energy out
Moon (in-distribution)	Adaptive risk-aware A*	50	0.92	0.829	0	3
	Distance-only A*	50	0.66	0.659	0	12
	Fixed risk-aware A*	50	0.90	0.788	0	4
Mars (OOD)	Adaptive risk-aware A*	50	0.34	0.158	7	20
	Distance-only A*	50	0.48	0.101	7	18
	Fixed risk-aware A*	50	0.34	0.173	6	21
Mars, 1.5x slip dispersion	Adaptive risk-aware A*	50	0.36	0.132	7	22
	Distance-only A*	50	0.44	0.081	8	20
	Fixed risk-aware A*	50	0.22	0.162	8	27
Mars, fault injection	Adaptive risk-aware A*	50	0.46	0.146	8	17
	Distance-only A*	50	0.40	0.074	9	20
	Fixed risk-aware A*	50	0.26	0.168	9	24
Mars, 20-step intervention delay	Adaptive risk-aware A*	50	0.44	0.156	2	23
	Distance-only A*	50	0.54	0.107	3	19
	Fixed risk-aware A*	50	0.28	0.200	4	26

Table 2: Primary contrasts: adaptive versus fixed risk-aware A*, paired within condition and seed. Holm-corrected across the primary family; * marks $p_{\text{Holm}} < 0.05$.

Condition	Outcome	Δ 95% CI	Test	p_{Holm}
Mars, 20-step intervention delay	Science fraction	-0.044 [-0.085, -0.002]	paired t-test	0.326
Mars, 20-step intervention delay	Mission success	+0.160 [+0.004, +0.316]	McNemar exact	0.461
Mars, fault injection	Science fraction	-0.022 [-0.050, +0.005]	paired t-test	0.461
Mars, fault injection	Mission success	+0.200 [+0.085, +0.315]	McNemar exact	0.020*
Mars, 1.5x slip dispersion	Science fraction	-0.030 [-0.059, -0.002]	paired t-test	0.326
Mars, 1.5x slip dispersion	Mission success	+0.140 [-0.001, +0.281]	McNemar exact	0.461
Mars (OOD)	Science fraction	-0.015 [-0.051, +0.022]	paired t-test	1.000
Mars (OOD)	Mission success	+0.000 [-0.128, +0.128]	McNemar exact	1.000
Moon (in-distribution)	Science fraction	+0.040 [+0.002, +0.079]	paired t-test	0.326
Moon (in-distribution)	Mission success	+0.020 [-0.071, +0.111]	McNemar exact	1.000

Table 3: Generalization gap in science fraction between the in-distribution lunar condition and the out-of-distribution Martian condition. See Section 6 for why this ranking is not informative.

Planner	In-distribution	Out-of-distribution	Gap G
Adaptive risk-aware A*	0.829	0.158	0.670
Distance-only A*	0.659	0.101	0.558
Fixed risk-aware A*	0.788	0.173	0.615

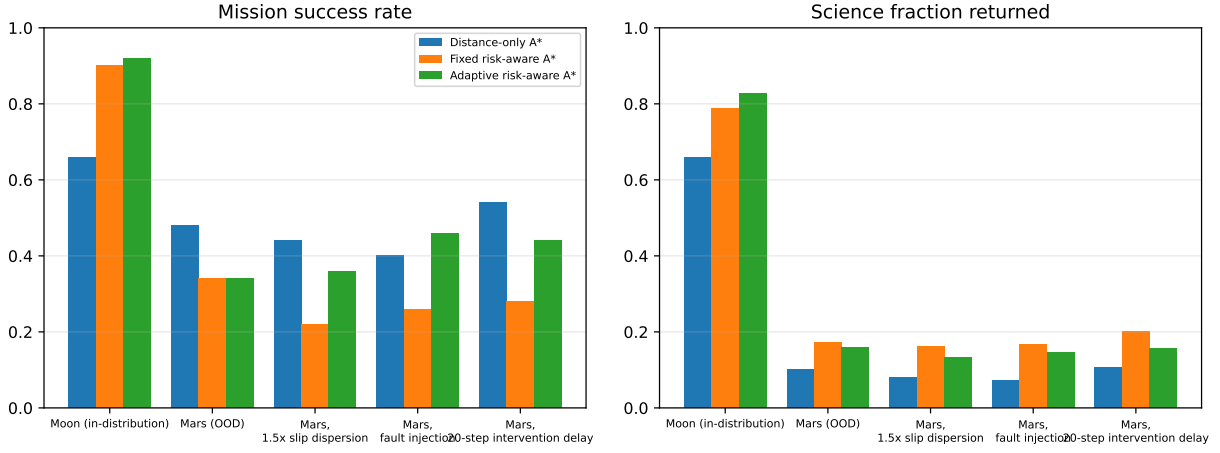


Figure 1: Mission success rate and science fraction by condition and planner.

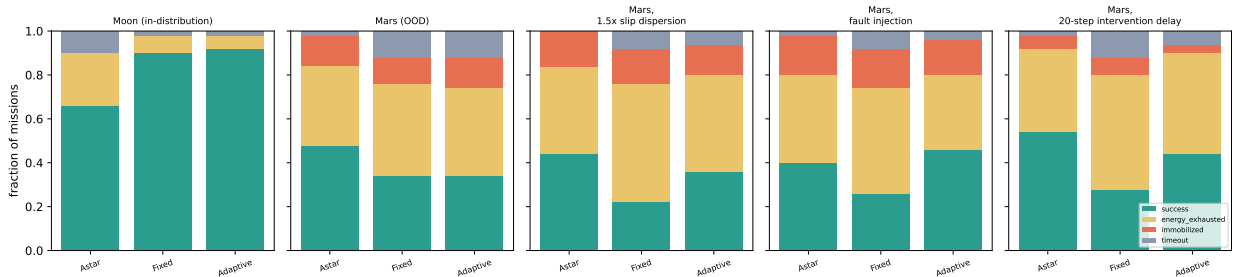


Figure 2: How missions ended. The effect of adaptation appears as a shift between termination causes – fewer energy exhaustions – rather than as a uniform improvement.

7 Discussion

The result we set out to find is not the result we obtained. H2, the hypothesis that online adaptation would raise science return under domain shift, is not supported: the effect was absent on held-out seeds despite appearing clearly during tuning. What survived confirmatory analysis is narrower and different in kind — adaptation improved *survival* under hardware faults, and did so by making the robot’s energy predictions honest enough that it turned back in time.

Three things follow.

First, the value of online adaptation in this testbed is concentrated where the robot’s own state is degrading, not merely where the terrain is unfamiliar. Faults change the vehicle underneath the model; a model that cannot be revised keeps planning for a rover that no longer exists.

Second, caution has a price that risk-aware formulations do not currently charge. On terrain where most ground is hazardous, detouring is not free, and the uninformed planner’s survival advantage came from simply spending less time exposed. A more complete objective would price route length and exposure time against per-cell risk rather than treating risk avoidance as costless.

Third, the gap between the validation and confirmatory estimates is the most methodologically useful outcome here. A tuning-set advantage of +0.125 to +0.167 in Martian success became 0.000 on held-out seeds. Had the seed protocol not been frozen in advance, that first number is the one that would have been reported.

7.1 Future work

The obvious extensions are to adapt the locomotion multiplier as well as the slip model, to add exposure time to the planner’s objective so that caution is priced, and to test transfer in the reverse direction (Mars prior, lunar deployment) to check that the effects here are not specific to one direction of shift.

8 Limitations

Stated plainly, because the value of this study depends on what it does not claim.

1. **This is not a validated rover model.** Terrain parameters were chosen to produce the qualitative regimes the experiment needs — a low-traction trap, a high-traction refuge, impassable rims, and true darkness — and are not calibrated against measured lunar or Martian geotechnical data. Absolute values are properties of this simulator.
2. **2.5-D, not dynamics.** There is no vehicle suspension, no wheel contact mechanics, and no attitude. Slip is drawn from a distribution rather than emerging from terramechanics.
3. **Risk composition is approximate.** Equations 6 and 7 treat per-cell events as independent, which understates risk in homogeneous terrain. It is applied identically to all planners.
4. **Only the slip model adapts.** The per-class locomotion multiplier is not revised online. Energy estimates still improve through the slip channel, since slip enters Equation 2 directly, but a systematic error in m_k would persist.
5. **The analysis plan is not a true pre-registration.** It was written after the adaptive planner existed and after an eight-seed engineering pilot had been seen. Those eight seeds are quarantined, and the full timeline is recorded in `docs/RESEARCH_LOG.md`.

6. **Two bodies, one prior direction.** Only Moon→Mars transfer is tested. The reverse direction, and bodies beyond these two, are untested.

9 Reproducibility

Every number above is generated by `scripts/make_exonaut_paper_assets.py` from the committed row-level CSV; no value is typed into this document. The run’s metadata sidecar records the git commit, the seed-split checksum, the quarantine list, and library versions. The full pipeline is:

```
python scripts/freeze_seed_protocol.py           # once; refuses to overwrite
python scripts/calibrate_priors.py               # priors from TRAIN seeds only
python scripts/run_exonaut_experiments.py        # confirmatory sweep
python scripts/make_exonaut_paper_assets.py      # tables, figures, macros
```

References

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